

FED-CAST: Federated Training and Evaluation Workflow for Generative Precipitation Nowcasting Across Climate-Diverse Regions

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Abstract—Short-term precipitation nowcasting is highly resource-intensive, requiring massive, climate-diverse radar archives for training. Aggregating this data at a single site creates severe communication overhead, making centralized training increasingly impractical. To address this problem, we present FED-CAST, a federated training and evaluation workflow for generative precipitation nowcasting. Instead of moving raw radar archives, FED-CAST trains directly at regional data sources and synchronizes only lightweight model parameters. We validate FED-CAST using the National Oceanic and Atmospheric Administration (NOAA) Multi-Radar/Multi-Sensor System (MRMS) Precipitation Rate product, partitioned across seven climate-diverse regional clients. Each client trains a Deep Generative Model of Radar (DGMR) nowcaster locally, and a central server synchronizes the model using Federated Averaging (FedAvg) with uniform client weighting.

To ensure reproducible comparison, FED-CAST integrates the Model-Compare-Tool (MCT). This provides a standardized pipeline for inference, event-driven benchmarking, multi-metric verification, TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) scoring, and MET/MODE object analysis. We compare FED-CAST against both a centralized DGMR and the Short-Term Ensemble Prediction System (STEPS) baseline using 1 to 48 months of training data. Under a fixed training budget, FED-CAST outperforms the centralized approach for 1–24 month training intervals. Specifically, at 1 month, FED-CAST achieves a TOPSIS score of 0.8637 (compared to 0.7019 centrally), yielding a peak instance-level advantage of 57.53%. As data scales to 48 months, the federated approach remains highly competitive, performing within $\sim 0.56\%$ of the centralized model. These results show that federated generative nowcasting successfully mitigates data movement overhead without sacrificing predictive skill, while providing a robust framework for exploring regional heterogeneity and standardized verification in data-intensive atmospheric modeling.

Index Terms—Federated Learning, Precipitation Nowcasting, Generative Forecasting, DGMR, Workflow, MRMS

I. INTRODUCTION

Extreme and rapidly evolving precipitation events impose substantial societal and economic costs, from loss of life and property in flash floods to cascading disruptions across transportation, energy, agriculture, and emergency services. High-resolution precipitation nowcasting, typically targeting 0–2 h lead times, can reduce these impacts by enabling earlier warnings, better allocation of response resources, and safer operations in weather-sensitive sectors [21]. Despite improvements in observational networks and forecasting methods,

actionable short-lead guidance remains difficult, especially for high-impact convective events whose location, intensity, and structure can evolve rapidly. In this paper, we frame the development of a precipitation nowcasting system as a data-intensive scientific workflow encompassing data management, distributed training orchestration, model-update communication, checkpointing, and standardized evaluation. The central question is whether generative nowcasting can be trained and evaluated across climate-diverse regional data sources while reducing the need to aggregate raw observations at a centralized site and preserving reproducible verification methodology.

Radar-based precipitation nowcasting spans classical extrapolation, deterministic neural sequence prediction, and probabilistic generative forecasting [1, 21]. Representative deep-learning models include convolutional long short-term memory (ConvLSTM) [27], Trajectory Gated Recurrent Unit (TrajGRU) [28], and the Deep Generative Model of Radar (DGMR) [23]. More recent generative approaches, including latent and physically guided diffusion models, further illustrate the growing role of probabilistic nowcasting [8, 13]. These models are commonly trained using centralized pipelines that assume large radar archives can be transferred, stored, and processed at a single compute site. This assumption motivates a workflow-oriented approach that integrates distributed data organization, training orchestration, and verification.

Fully centralized training faces practical constraints from the volume and velocity of operational radar streams. The Next Generation Weather Radar (NEXRAD) network consists of 160 high-resolution S-band Doppler radars, and the National Centers for Environmental Information (NCEI) archives Level-II files containing 4.5 to 10 minutes of base data depending on the volume coverage pattern [16]. A single active radar can produce approximately 0.6–1.0 GB/h of raw observations before downstream preprocessing [30]. Scaling ingestion from dozens or hundreds of radars simultaneously creates substantial bandwidth, storage, and operational overhead. Beyond data volume, institutional data-governance constraints and the risk of single-point-of-failure architectures further motivate distributed alternatives.

To address these challenges, we develop FED-CAST, a federated training and evaluation workflow for generative precipitation nowcasting. Federated learning (FL) [15] enables

each site to train on its local data and share only model updates with a central server, which aggregates them using federated averaging (FedAvg). This design naturally aligns with the structure of radar precipitation observations: each site or regional subdomain learns from its own precipitation regime while contributing to a shared generative nowcasting model, without requiring centralized transfer of large multi-frame spatiotemporal tensors. Recent federated weather-modeling studies have explored region-specific adaptation for radar nowcasting and communication-efficient prompt exchange for broader weather forecasting [4, 25]. FED-CAST focuses on a complementary integration problem: pairing probabilistic generative nowcasting with regionally heterogeneous data sources and a fixed event-driven verification framework.

We select seven radar-centered regional domains representing climatologically distinct U.S. precipitation regimes as distributed clients in FED-CAST. Each client trains a DGMR-based generative nowcast model locally, while a federated orchestration layer [2] coordinates FedAvg-style aggregation with uniform client weighting at the server. We refer to the resulting model as *federated DGMR*; FED-CAST denotes the broader workflow within which it is trained and evaluated. This setup provides a controlled scenario for studying federated generative nowcasting under heterogeneous data distributions.

The workflow includes a standardized evaluation component, the Model-Compare-Tool (MCT), which serves as a unified framework for comparing centralized DGMR, federated DGMR, and the Short-Term Ensemble Prediction System (STEPS) baseline implemented through the open-source PySTEPS library [3, 22, 26]. MCT standardizes model inference, output formatting, event-driven benchmark evaluation, deterministic and probabilistic verification, and multi-metric summarization using the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS). Additional baselines, including NowcastNet and TrajGRU [31], are incorporated for centralized-training comparisons; implementation details and the integration interface are described in Sec. IV-D.

Our experimental study uses this workflow to compare centralized and federated DGMR across six training-data intervals from 1 to 48 months, with STEPS as a classical probabilistic baseline. The experiment evaluates whether the proposed workflow supports controlled and reproducible comparisons under matched data-interval and training-budget constraints, and examines the data scale required for the models to achieve acceptable and stable performance. We hypothesize that federated DGMR can match centralized DGMR overall and be more data-efficient in smaller training-data settings, given that each client specializes in a distinct precipitation regime; thereby emphasizing the feasibility of the proposed federated training-and-evaluation workflow. We also include controlled ablations on client-aggregation weighting and Sharpness-Aware Minimization (SAM) [7] to probe whether explicit regularization improves small-dataset learning behavior.

In evaluating this hypothesis, we make the following main contributions in this paper:

- **FED-CAST workflow:** a Flower-based federated pipeline connecting data preparation, local DGMR training, FedAvg-style aggregation with uniform client weighting, checkpointing, and a standard performance evaluation.
- **Climate-diverse client testbed:** seven MRMS-derived regional clients representing diverse U.S. precipitation regimes, serving as naturally distributed data sources for federated training.
- **Unified MCT evaluation:** a shared event-driven benchmark for centralized DGMR, federated DGMR, and STEPS, with categorical, probabilistic, quality-control, runtime, TOPSIS, and object-based verification.
- **Controlled ablations:** aggregation-weighting and explicit-SAM analyses that reveal the conditions under which regularization and weighting schemes benefit small-dataset learning, supplementing the primary workflow contribution.
- **Empirical feasibility evidence:** under a fixed training budget across 1–48 month data intervals, FED-CAST outperforms centralized DGMR for 1–24 month training periods, achieving a peak instance-level TOPSIS advantage of 57.53% at 1 month (0.8637 vs. 0.7019). At 48 months, federated DGMR remains within $\sim 0.56\%$ of centralized performance, demonstrating that distributed training mitigates data-movement overhead without sacrificing predictive skill.

II. BACKGROUND

A. Precipitation Nowcasting and MRMS Data

Precipitation nowcasting focuses on forecasting the evolution of rainfall within the next minutes to two hours. Radar-derived observations provide the high spatial and temporal resolution needed for this task. For distributed nowcasting studies, the data source must support both regional heterogeneity and cross-region consistency in the U.S. The NOAA Multi-Radar Multi-Sensor (MRMS) system integrates observations from the nationwide Weather Surveillance Radar–1988 Doppler (WSR-88D) network, satellites, rain gauges, and numerical weather prediction fields to produce quality-controlled, gridded precipitation products [32]. Because MRMS integrates observations across sensors and regions, it provides a suitable foundation for constructing regional subdomains that emulate geographically distributed radar data sources while keeping the model input format consistent across clients.

B. Generative Nowcasting with DGMR

Deep-learning nowcasting models learn the temporal evolution of precipitation fields directly from recent radar-derived sequences. DGMR [23], is a conditional generative adversarial model designed to sample plausible future precipitation fields from past observations. This generative formulation is suitable for short-term precipitation forecasting because convective evolution is uncertain and can have multiple physically plausible futures. In this study, DGMR serves as the common neural nowcasting model for both centralized and federated

training, allowing the training workflow to vary while keeping the model class fixed.

C. Federated Learning and Regional Heterogeneity

Federated learning (FL) enables decentralized model training across multiple data sources by sharing model parameters or model updates rather than raw data. In the standard **FedAvg** [15] algorithm, a central server broadcasts the current global model to participating clients; each client performs local optimization; and the server aggregates the resulting client models, commonly using sample-size-proportional weights. This paradigm is relevant for precipitation nowcasting because radar-derived data are large, frequently updated, and naturally organized by geographic sources. In this work, MRMS spatial partitions provide a controlled way to study this setting: each regional subdomain acts as a model training participant with its own precipitation distribution, while all clients share the same tensor format, model architecture, and training protocol.

We use an equal-weighted FedAvg-style aggregation rule as the primary client-balanced workflow baseline. This choice is intended to prevent regions with more sequences remaining after preprocessing from dominating the federated aggregation and allows the primary experiments to focus on decentralized training under climate-driven heterogeneity. We define the specific server-side weighting rules for the baseline and ablation experiments in Sec. IV-C and Sec. V-C.

D. Nowcast Verification Metrics

Precipitation nowcasts are typically evaluated with complementary metrics because no single score captures all forecast properties. Threshold-based categorical scores, such as critical success index (CSI), probability of detection (POD), false alarm ratio (FAR), Heidke skill score (HSS), and bias, measure event detection skill; probabilistic scores such as continuous ranked probability score (CRPS) assess ensemble forecast distributions; and continuous or signal-fidelity metrics such as mean absolute error (MAE), root mean squared error (RMSE), and peak signal-to-noise ratio (PSNR) assess field-level reconstruction quality. Open-source tools such as PySTEPS provide categorical and pixel-wise probabilistic verification utilities for precipitation nowcasting [22], while Model Evaluation Tools (MET and METplus) provide a broader verification framework for atmospheric prediction [29]. Pixel-wise metrics can double-penalize spatially coherent forecasts that are slightly displaced, motivating object-based methods such as MODE, which compare precipitation objects using location, size, orientation, and overlap attributes [5, 6].

III. DATA SOURCES AND PREPROCESSING

A. MRMS precipitation products

We use the NOAA MRMS `PrecipRate` product, a gridded precipitation-rate estimate on a 0.01° grid with nominal 2-min cadence [19, 24, 32].

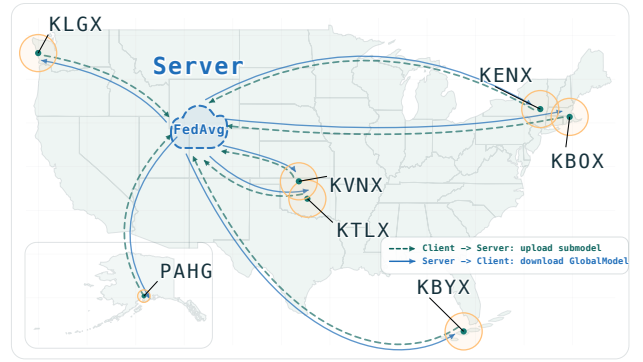


Fig. 1. Geographic locations of the seven radar-centered regional subdomains used as federated clients. The cloud-shaped server icon is schematic and denotes a cloud-side role, not an actual deployment location.

B. Radar-centric spatial subsetting

To emulate geographically distributed data sources, as shown in Fig.1, we extract seven radar-centered subdomains from MRMS using Weather Surveillance Radar–1988 Doppler (WSR-88D) locations [17]. These seven sites were selected to provide broad and complementary coverage of major U.S. precipitation regimes while retaining sufficient precipitation-rich sequences, spanning subtropical maritime precipitation near the Florida Keys, continental convective regimes over the Great Plains, Pacific coastal and orographic precipitation, inland and coastal Northeast mid-latitude regimes, and high-latitude Alaskan precipitation. Each client dataset is based on a $3^\circ \times 3^\circ$ area around a radar, yielding a 300×300 field at MRMS resolution.

As presented in Fig.2, each output is a 16-frame sequence at 2-min cadence. We use 4 frames as input and 12 frames as targets (we pick 3 frames from the targets to demonstrate the example output), so every model receives an 8-min context window and predicts the next 24 min.

C. Train/validation/test split

We use an approximately 80/10/10 train/validation/test split for model development. The training split is used to update model parameters, the validation split is used for checkpoint selection, and the test split is held out from both training and checkpoint selection.

To construct the test split, we reserve the first three available days of each month as contiguous held-out calendar blocks. This yields an approximately 10% test fraction while ensuring that the test set spans all months and seasons. The first-three-day rule has no meteorological significance; it is used as a deterministic and reproducible monthly hold-out scheme.

From the remaining pool, we sample a validation split of comparable size using a fixed random seed. This validation set is used to select the final checkpoint, specifically the model checkpoint with the lowest validation loss. It is distinct from the test split, which is not used for training, checkpoint selection, or model tuning. All remaining sequences are used for training. Split manifests are stored and reused so that

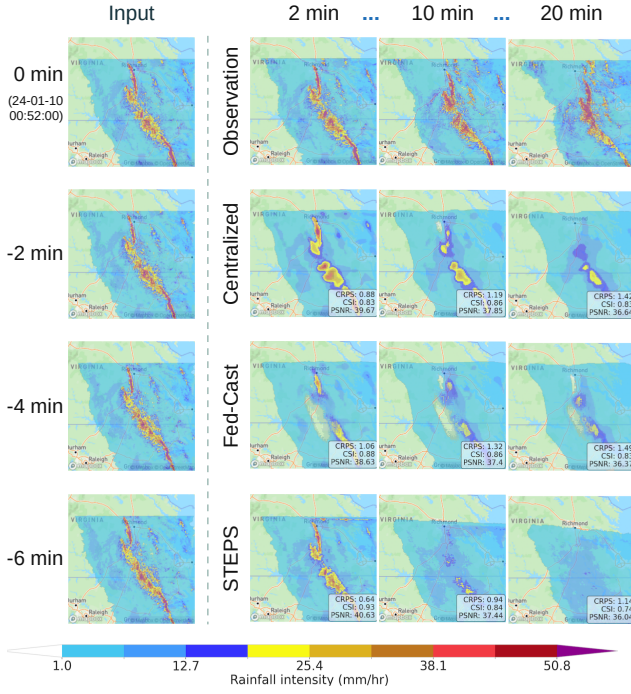


Fig. 2. Qualitative nowcasting example for one event. The leftmost column shows the four observed input frames shared by all methods. The remaining columns show selected target and forecast lead times at 2, 10, and 20 min. Rows show, from top to bottom, the actual observations, centralized DGMR, federated DGMR, and STEPS. Intermediate lead times at 4, 6, 8, 12, 14, 16, and 18 min are omitted for space but are included in the quantitative evaluation. Color indicates rainfall intensity in mm h^{-1} .

centralized and federated runs use identical train, validation, and test partitions.

D. Event-driven benchmark for standardized evaluation

All three approaches (centralized DGMR, FED-CAST, and STEPS) are evaluated on a fixed benchmark, which is composed of weather events selected from the following databases:

- Weather Prediction Center (WPC) Mesoscale Precipitation Discussions (MPD) [12];
- Local Storm Reports (LSR) [11];
- NOAA/NCEI Storm Events Database [18].

All events are standardized into a unified event table containing: event ID, source, time window in Coordinated Universal Time (UTC), and a geographic footprint.

IV. METHODOLOGY

A. Problem setup and workflow design

For the evaluation of the different nowcasting approaches, we consider a scenario with 2-minute rainfall rate updates. For each nowcast, four observations at $t - 8\text{min}$, $t - 6\text{min}$, $t - 4\text{min}$, $t - 2\text{min}$ serve as input to produce 12 forecasts at 2-minute intervals from t to $t + 24\text{min}$.

Figure 3 summarizes the workflow that connects MRMS regional subsetting, radar-centered client data preparation, federated DGMR training, checkpoint selection, and standardized verification through our Model-Compare-Tool (MCT).

We train models using six training-data intervals $L \in \{1, 3, 6, 12, 24, 48\}$ months, while keeping the data preprocessing, input/output definition, model architecture, training budget, checkpoint rule, and event-driven evaluation benchmark fixed. This design allows us to compare training paradigms under matched conditions and to examine how the amount of historical training data affects each paradigm. At this design level, centralized DGMR uses 100 training epochs, while FED-CAST uses 100 communication rounds with all seven clients participating and one local epoch per client per round; the detailed training and checkpointing protocols are specified in Sec. IV-C.

We compare the following methods:

- **DGMR (centralized):** a DGMR-based generative nowcasting model trained on pooled data from all seven regional subdomains [23].
- **DGMR (federated):** the same DGMR model trained across seven radar-centered clients using a Flower-orchestrated FedAvg-style protocol with uniform client weighting [2, 15].
- **STEPS baseline:** a classical stochastic nowcasting baseline implemented through PySTEPS.

To ensure a fair and reproducible comparison between model classes and training regimes, every trained model is evaluated on the same fixed set of event-driven benchmarks (Sec. III-D) using MCT.

B. Forecasting models

1) *DGMR stochastic ensemble generation:* We adopt DGMR as the core generative nowcasting model [23]. In our configuration, DGMR predicts 12 lead times (`forecast_steps=12`). During inference and evaluation, we generate an ensemble of $K = 6$ stochastic forecasts per input sequence (`num_samples=6`) by running the generator six times with independent stochastic inputs.

For an input sequence $\mathbf{x}$, the k -th stochastic DGMR forecast is written as

$$\hat{\mathbf{y}}^{(k)} = f_{\theta}(\mathbf{x}; \mathbf{z}^{(k)}), \quad k = 1, \dots, K, \quad (1)$$

where $\mathbf{z}^{(k)}$ denotes the stochastic latent or noise input used in the k -th forward pass. If the input for a particular indicator requires a deterministic forecast rather than an ensemble forecast, MCT uses the ensemble mean

$$\bar{\mathbf{y}} = \frac{1}{K} \sum_{k=1}^K \hat{\mathbf{y}}^{(k)}, \quad K = 6. \quad (2)$$

Probabilistic metrics, such as CRPS, use the full six-member DGMR ensemble. The same DGMR architecture and hyperparameter configuration are used for both centralized and federated training; only the training workflow differs.

2) *PySTEPS STEPS baseline:* As a strong classical baseline, we run STEPS using PySTEPS as a scale-decomposition stochastic nowcasting scheme. Given the input sequence and an estimated motion field, we generate a 20-member ensemble for 12 lead times with six cascade levels, nonparametric

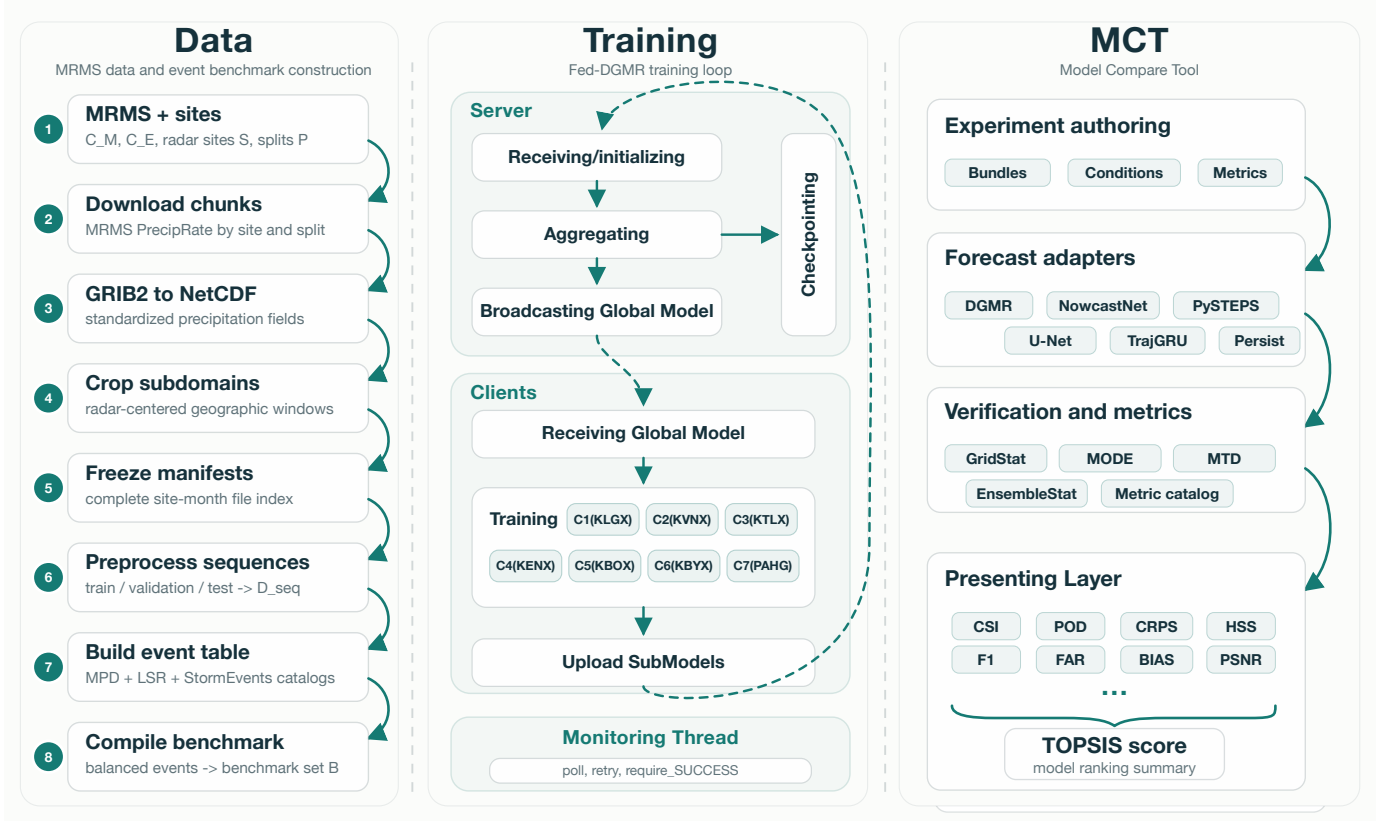


Fig. 3. FED-CAST workflow linking MRMS radar-centered data construction, federated DGMR training, and MCT-based multi-metric evaluation.

precipitation noise, Bowler–Pierce–Seed (BPS) velocity perturbations, and an incremental precipitation mask.

C. Training Strategies

1) *Centralized DGMR training*: Centralized DGMR training is implemented with PyTorch Lightning on a graphics processing unit (GPU). For each training-data interval L , the centralized model is trained on the combined training sequences from all seven regional subdomains. We train for 100 epochs and run validation every 5 epochs. We save the model checkpoints and select the candidate model with the lowest **generator validation loss** as the final model for evaluation. Here, **generator validation loss** is the logged DGMR generator objective,

$$L_{\text{val},G} = -\text{mean}_{k=1}^6 D([\mathbf{x}, G_k(\mathbf{x})]) + \lambda_{\text{grid}} \mathcal{L}_{\text{grid}}(\bar{\mathbf{y}}, \mathbf{y}), \quad (3)$$

$$\bar{\mathbf{y}} = \frac{1}{6} \sum_{k=1}^6 G_k(\mathbf{x}).$$

Here $G_k(\mathbf{x})$ is the k -th stochastic DGMR forecast, $D(\cdot)$ is the discriminator score used by the generator hinge loss, $\mathbf{y}$ is the target future sequence, $\mathcal{L}_{\text{grid}}$ is the grid-cell regularizer, and $\lambda_{\text{grid}} = 20$ in our configuration, matching the Open Climate Fix DGMR implementation [20].

All unspecified trainer settings use PyTorch Lightning defaults. DGMR hyperparameters not explicitly overridden follow the referenced DGMR implementation.

2) *Federated DGMR training with Flower*: Federated training is performed with Flower [2] using a synchronous client-server FedAvg-style protocol [15]. We define seven clients, each corresponding to one radar-centered regional subdomain (Sec. III-B). In each FL round, **all seven clients participate**. Each client receives the current global model, runs **one local epoch** on its local training sequences, and returns the post-local-training model to the server.

The baseline federated workflow applies uniform client weighting: each client is assigned the same server-side aggregation coefficient. Thus, the baseline FL update is

$$\theta^{(t+1)} = \frac{1}{7} \sum_{i=1}^7 \theta_i^{(t+1)}, \quad (4)$$

where $\theta_i^{(t+1)}$ denotes the model returned by client i after local training in round t . This baseline does not use sample-size-proportional or retained-sequence-count-proportional weighting at the server. This uniform-weight baseline assigns the same aggregation coefficient to each client, regardless of its remaining sequences. This prevents sample-rich regions from dominating the primary comparison, while client-size-aware weighting rules are evaluated separately in Sec. V-C.

We run federated training for 100 rounds, which is aligned with the 100 epochs used in centralized training. Validation is performed every 5 rounds, and the global model checkpoint with the lowest **generator validation loss** is selected as the final model for evaluation.

TABLE I
METRIC SUITE USED FOR TOPSIS AGGREGATION IN THE REPORTED EXPERIMENTS.

Metric	Pref.	Full name
POD	↑	Probability of detection
PSNR	↑	Peak signal-to-noise ratio
ACC	↑	Accuracy
CSI	↑	Critical success index
HSS	↑	Heidke skill score
GSS	↑	Gilbert skill score
MCC	↑	Matthews correlation coefficient
F1	↑	F1 score
SEDI	↑	Symmetric extreme dependency index
RAPSD	↓	Radially averaged power spectral density distance
FAR	↓	False alarm ratio
FA	↓	False alarm rate
CRPS	↓	Continuous ranked probability score
HK	↑	Hanssen–Kuipers skill score
BIAS	→ 1	Event-frequency bias
Executing_Time	↓	Inference execution time

D. Standardized evaluation with Model-Compare-Tool (MCT)

We developed MCT to standardize evaluation across model classes, training paradigms, and ensemble formats. MCT acts as the common verification layer for centralized DGMR, federated DGMR, and STEPS. It provides:

- model-specific inference adapters that convert each method’s output into a common tensor format;
- evaluation on the fixed event-driven benchmark dataset (Sec. III-D);
- a unified metric interface implemented with PySTEPS, TorchMetrics, and the enhanced Model Evaluation Tools (METplus) verification system built around the core Model Evaluation Tools (MET) object-based verification modules [29].

MCT treats ensemble forecasts consistently across methods. DGMR produces a six-member ensemble, while STEPS produces a 20-member ensemble. For probabilistic verification, MCT evaluates the full ensemble distribution. For deterministic scores such as CSI, POD, and PSNR, MCT first converts each ensemble into an ensemble-mean deterministic forecast.

MCT also supports an extended object-based verification path through the Method for Object-Based Diagnostic Evaluation (MODE) in the METplus toolchain [5, 6]. This path compares spatially coherent precipitation objects rather than only gridpoint overlap.

E. Metrics and TOPSIS aggregation

MCT computes verification metrics spanning categorical skill scores, distributional and probabilistic scores, signal-fidelity metrics, spectral-distribution distances, and runtime. Table I lists the metric suite used for TOPSIS aggregation in the reported experiments.

Categorical metrics. Deterministic categorical scores, including CSI, POD, HSS, Gilbert skill score (GSS), Matthews correlation coefficient (MCC), F1 score, and symmetric extreme dependency index (SEDI), are computed from pixel-level contingency tables using PySTEPS verification utilities [22]. We use a precipitation threshold of $\theta = 0.1 \text{ mm h}^{-1}$ to define “event” and “non-event” pixels.

Lead-time handling. All metrics are computed *per lead time* and then averaged across the 12 lead times to obtain one summary value per metric, per model.

TOPSIS multi-metric summary. To summarize multi-metric performance into a single scalar for visualization and comparison, we apply the TOPSIS with **equal weights** across metrics [10]. TOPSIS ranks each method by its distance to an ideal best solution and an ideal worst solution under the chosen metric set, producing one closeness coefficient per model. Larger TOPSIS values indicate better overall performance under the selected metric inventory and preference directions. In the TOPSIS matrix, benefit metrics use the upward preference direction, cost metrics use the downward preference direction, and BIAS is treated as a target-to-one metric.

Additional controlled ablations that modify the server aggregation rule or centralized optimizer are introduced together with their results in Sec. V-C, after the primary centralized-versus-federated comparison. To make the experiment identifiers used in the results section explicit, Table II summarizes the reported experiment sequence.

TABLE II EXPERIMENT IDENTIFIERS USED IN THE REPORTED RESULT SEQUENCE.	
ID	Experiment role
E1	Primary comparison of centralized DGMR, FED-CAST, and STEPS across the six training-data intervals using the fixed event benchmark, MCT metric suite, and equal-weight TOPSIS summary.
E2.1	Aggregation-rule ablation that replaces FED-CAST’s uniform client weighting with quadratic count-derived client weights to test whether the small-dataset FL trend is mainly caused by equal-client averaging.
E2.2	Optimization ablation that adds generator-side explicit SAM to centralized DGMR while leaving the federated comparison unchanged. Here, ρ denotes the SAM perturbation radius, which sets the magnitude of the sharpness-aware weight perturbation before the second optimizer pass. The reported E2.2 panels combine two sensitivity settings, $\rho = 0.025$ and $\rho = 0.0125$.

V. EXPERIMENTS AND RESULTS

This section reports results under the design defined in Sec. IV-A and the training protocols in Sec. IV-C. In brief, centralized DGMR, FED-CAST, and STEPS are compared across six training-data intervals using the same event-driven benchmark dataset, MCT verification pipeline, and TOPSIS metrics. The goal is not to maximize the absolute performance of every model, but to compare how centralized and federated DGMR training behave under the same model training interval and compute constraints.

A. Aggregate TOPSIS Results

For the primary comparison (E1) (see Tab. II), Fig. 4 reports the equal-weight TOPSIS scores for centralized DGMR, FED-CAST, and STEPS across the six training-data intervals. For each DGMR-based method and training interval L , boxes summarize the date-tagged TOPSIS distribution using the interquartile range, median, and min–max whiskers, while trend lines connect the medians across L . Solid and dashed trend lines distinguish centralized and federated DGMR, and the dotted horizontal line gives the STEPS reference score. Table III summarizes the shared training-period end months and the resulting number of trained DGMR models per centralized

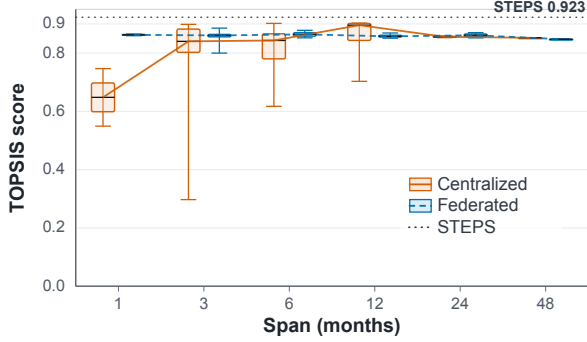


Fig. 4. E1 TOPSIS learning curves for centralized DGMR, federated DGMR, and STEPS across training-data intervals L on the event-driven benchmark.

or federated paradigm. Because data preprocessing filters out data that provide insufficient precipitation information, only four months out of the month-long data contain enough valid data for all radar stations.

STEPS is used as a stable classical reference rather than the primary optimization target. Its strong score should be read in that context: the DGMR-based models use a fixed feasibility-study budget, whereas prior DGMR work shows stronger performance with sufficient data and compute [23, 31].

Under this fixed training-epoch budget, FED-CAST outperforms centralized DGMR for training periods from 1 to 24 months. The largest gap appears in the 1-month setting, where FED-CAST achieves a TOPSIS score of 0.8637 compared with 0.7019 for centralized DGMR (see Fig. 4). FED-CAST remains stable across training spans (0.8476–0.8637), while centralized DGMR improves with more data and slightly exceeds FED-CAST at 48 months (0.8524 vs. 0.8476).

FED-CAST is also more stable across different training periods. Its TOPSIS scores remain in a narrow range from 0.8476 to 0.8637, while centralized DGMR varies more substantially across the same data spans. This pattern suggests that federated training provides a more data-efficient training regime under limited data and fixed compute budgets, while centralized training narrows the gap as the available training data increases.

TABLE III
USABLE TRAINING-PERIOD ENDS AND RESULTING TRAINED-MODEL COUNTS AFTER PREPROCESSING.

Span (mo)	Models per paradigm	Training-period ends (year: month)
1	2	2022: 02, 10
3	16	2022–2025: 01, 04, 07, 10
6	8	2022–2025: 04, 10
12	4	2022–2025: 10
24	2	2022, 2024: 10
48	1	2024: 10

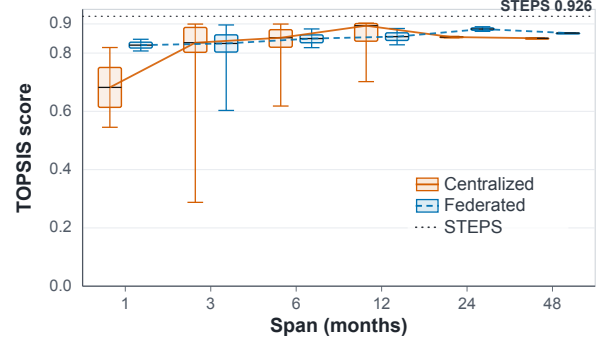


Fig. 5. E2.1 TOPSIS learning curves under quadratic count-derived aggregation. Boxes summarize date-tagged TOPSIS distributions across training windows L , and median trends compare the ablated federated workflow with centralized DGMR and the STEPS reference.

B. Interpretation of Centralized and Federated Trends

The results show that the federated workflow is not only feasible but also competitive under restricted training budgets. FED-CAST benefits most in the small training-data regime, where each client contributes region-specific precipitation examples and the global aggregation process combines information across climate-diverse subdomains. This behavior is valuable for data-intensive meteorological workflows because operational data can not always be pooled into a single centralized training repository.

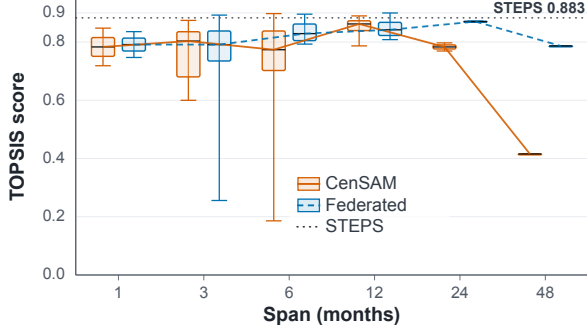
The centralized model follows a different pattern. Its aggregate performance improves as the available training data increases, which indicates that the pooled DGMR model can use larger training sets effectively. However, the centralized model requires sufficient data and compute before it reaches the same performance range as FED-CAST. This trend supports the use of federated training as a practical alternative when data movement, storage, governance, or compute constraints limit centralized training.

C. Controlled Ablation Analyses

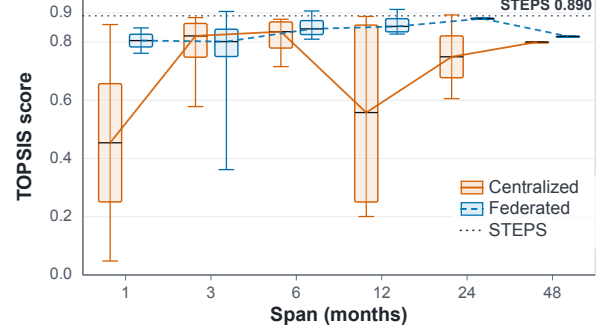
The primary results above show a small-dataset advantage for FED-CAST over centralized DGMR under the fixed training budget and reported MCT metric set. We therefore conduct two controlled ablations, E2.1 and E2.2, to test whether this behavior is mainly driven by the server aggregation rule or by centralized optimization characteristics. These analyses are secondary to the main workflow comparison and are used only to interpret the observed results.

a) *Aggregation-rule ablation (E2.1).*: The baseline FED-CAST workflow uses uniform client weighting, where each of the seven clients receives the same server-side aggregation coefficient.

Because preprocessing filters out data with insufficient precipitation information, data set sizes differ across clients. In this scenario, clients with smaller local datasets impact the global model as much as clients with larger local datasets. Here, we replace this baseline with a stronger client-size-aware



(a) $\rho = 0.025$



(b) $\rho = 0.0125$

Fig. 6. E2.2 TOPSIS learning curves for centralized DGMR with generator-side SAM at $\rho = 0.025$ and $\rho = 0.0125$, compared with the unchanged federated workflow and STEPS. Boxes summarize date-tagged TOPSIS distributions across L , and median trends indicate the sensitivity of the centralized trajectory to SAM strength.

aggregation rule. For client i , let n_i be the number of retained preprocessed training sequences and $n_{\max} = \max_{j=1,\dots,7} n_j$. We define

$$w_i = \max(1, \lfloor n_i^2 / n_{\max} \rfloor), \quad \alpha_i = \frac{w_i}{\sum_{j=1}^7 w_j}, \quad \theta^{(t+1)} = \sum_{i=1}^7 \alpha_i \theta_i^{(t+1)} \quad (5)$$

This rule is called quadratic because n_i appears twice in the unnormalized weight, giving clients with more data stronger-than-linear influence before normalization. As shown in Fig. 5, it is not intended to reproduce centralized learning; a linear rule $\alpha_i \propto n_i$ would be closer to centralized sample-frequency weighting. Instead, this approach is a stronger-than-linear test of whether the small-dataset FED-CAST advantage is mainly caused by uniform client weighting. The results show that this aggregation variant weakens but does not remove the small-dataset advantage, suggesting that uniform client weighting alone is not a sufficient explanation of why federated trained models perform better in the smaller training-data cases.

b) Optimization ablation (E2.2): To evaluate whether the small-dataset behavior can be partially reproduced without federation, we add generator-side Sharpness-Aware Minimization (SAM) [7] to centralized DGMR while keeping the architecture, losses, validation rule, training budget, and discriminator/generator update schedule unchanged. In this analysis, ρ is the SAM perturbation radius: larger values impose a stronger local weight perturbation when probing sharpness before the update. Fig. 6 compares the primary $\rho = 0.025$ setting with a lower-strength sensitivity run at $\rho = 0.0125$. This ablation is motivated by related analyses of local, decentralized, and federated optimization dynamics [9, 14, 33], but should not be read as evidence that FedAvg is equivalent to SAM. Across the two perturbation radii, explicit SAM can shift centralized DGMR toward the federated trend in shorter training periods, while the response remains sensitive to the SAM strength and less stable at longer periods.

Together, these analyses narrow the interpretation of the FED-CAST trend. The aggregation-rule evaluation shows that equal client weighting is not sufficient, while the SAM ablation

indicates that centralized sharpness-aware optimization can reproduce part of the small-data behavior. Overall, FED-CAST enables controlled centralized–federated DGMR comparison across climate-diverse MRMS clients: under the fixed training budget, it is stronger and more stable for 1–24 month windows, while centralized DGMR narrows the gap with scale and STEPS remains a stable classical reference rather than the main optimization target.

VI. CONCLUSION & FUTURE WORK

Short-term precipitation nowcasting relies on large radar-derived spatiotemporal fields whose statistics vary across regional climate regimes, making repeated wide-area data transfer to a central training site costly or impractical. In this setting, FED-CAST uses seven MRMS-derived radar-centered clients to compare federated DGMR, centralized DGMR, and STEPS on a common event-driven MCT benchmark spanning 1–48 months of training data. FED-CAST demonstrates a distributed training-and-evaluation workflow for generative precipitation nowcasting under climate-driven data heterogeneity. It emulates seven MRMS-derived regional clients, trains a shared DGMR-based nowcaster through FedAvg-style aggregation with uniform client weighting, and evaluates centralized DGMR, federated DGMR, and STEPS through the same event-driven MCT benchmark. Under the reported metric inventory, FED-CAST outperforms centralized DGMR for $L \in \{1, 3, 6, 12, 24\}$, where L is in months, and remains near parity at $L = 48$, improves CSI and POD relative to STEPS, but trails STEPS in CRPS and therefore in equal-weight TOPSIS. The ablations show that quadratic client reweighting weakens but does not eliminate the small-data advantage, while SAM moves centralized DGMR toward the federated trend at short windows with reduced stability at longer windows. Next, we will refine aggregation and SAM settings before rerunning the completed experiments with the MET/MODE-enabled MCT suite.

REFERENCES

- [1] Sojung An, Tae-Jin Oh, Eunha Sohn, and Donghyun Kim. 2025. Deep learning for precipitation nowcasting: A survey from the perspective of time series forecasting. *Expert Systems with Applications* 268 (2025), 126301. doi:10.1016/j.eswa.2024.126301
- [2] Daniel J. Beutel, Taner Topal, Akhil Mathur, et al. 2020. Flower: A Friendly Federated Learning Research Framework. *arXiv preprint arXiv:2007.14390* (2020).
- [3] Nick E. Bowler, Chris E. Pierce, and Alan W. Seed. 2006. STEPS: A probabilistic precipitation forecasting scheme which merges an extrapolation nowcast with downscaled NWP. *Quarterly Journal of the Royal Meteorological Society* 132, 620 (2006), 2127–2155. doi:10.1256/qj.04.100
- [4] Shengchao Chen, Guodong Long, Tao Shen, and Jing Jiang. 2023. Prompt Federated Learning for Weather Forecasting: Toward Foundation Models on Meteorological Data. In *Proceedings of the 32nd International Joint Conference on Artificial Intelligence (IJCAI)*. 3532–3540. doi:10.24963/ijcai.2023/393
- [5] Christopher A. Davis, Barbara G. Brown, and Randy Bullock. 2006. Object-Based Verification of Precipitation Forecasts. Part I: Methodology and Application to Mesoscale Rain Areas. *Monthly Weather Review* 134, 7 (2006), 1772–1784. doi:10.1175/MWR3145.1
- [6] Christopher A. Davis, Barbara G. Brown, Randy Bullock, and John E. Halley Gotway. 2009. The Method for Object-Based Diagnostic Evaluation (MODE) Applied to Numerical Forecasts from the 2005 NSSL/SPC Spring Program. *Weather and Forecasting* 24, 5 (2009), 1252–1267. doi:10.1175/2009WAF222241.1
- [7] Pierre Foret, Ariel Kleiner, Hossein Mobahi, and Behnam Neyshabur. 2021. Sharpness-Aware Minimization for Efficiently Improving Generalization. In *International Conference on Learning Representations (ICLR)*.
- [8] Zhihan Gao, Xingjian Shi, Boran Han, Hao Wang, Xiaoyong Jin, Danielle Maddix, Yi Zhu, Mu Li, and Yuyang Wang. 2023. PreDiff: Precipitation Nowcasting with Latent Diffusion Models. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [9] Xinran Gu, Kaifeng Lyu, Longbo Huang, and Sanjeev Arora. 2023. Why (and When) does Local SGD Generalize Better than SGD?. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=svCcu6Drl>
- [10] Ching-Lai Hwang and Kwangsun Yoon. 1981. *Multiple Attribute Decision Making: Methods and Applications*. Springer-Verlag.
- [11] Iowa Environmental Mesonet (IEM). 2026. Local Storm Reports (LSR) GeoJSON Service. <https://mesonet.agron.iastate.edu/geojson>. Accessed 2026-01-05.
- [12] Iowa Environmental Mesonet (IEM). 2026. WPC Mesoscale Precipitation Discussion (MPD) GIS Request Interface. https://mesonet.agron.iastate.edu/cgi-bin/request/gis/wpc_mpd.py?help=. Accessed 2026-01-05.
- [13] Jussi Leinonen, Ulrich Hamann, et al. 2023. Latent diffusion models for generative precipitation nowcasting with accurate uncertainty quantification. *arXiv preprint arXiv:2304.12891* (2023). doi:10.48550/arXiv.2304.12891
- [14] Jinwoo Lim, Suhyun Kim, and Soo-Mook Moon. 2025. Convergence Analysis of Federated Learning Methods Using Backward Error Analysis. *Proceedings of the AAAI Conference on Artificial Intelligence* 39, 18 (2025), 18719–18727. doi:10.1609/aaai.v39i18.34060
- [15] H. Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. 2017. Communication-Efficient Learning of Deep Networks from Decentralized Data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*.
- [16] NOAA National Centers for Environmental Information. [n.d.]. Next Generation Weather Radar (NEXRAD). <https://www.ncei.noaa.gov/products/radar/next-generation-weather-radar>. Accessed: Apr. 28, 2026.
- [17] NOAA National Centers for Environmental Information (NCEI). 2026. NEXRAD Station List. <https://www.ncei.noaa.gov/access/homr/file/nexrad-stations.txt>. Accessed 2026-01-05.
- [18] NOAA National Centers for Environmental Information (NCEI). 2026. Storm Events Database (StormEvents_details CSV files). <https://www.ncei.noaa.gov/pub/data/swdi/stormevents/csvfiles/>. Accessed 2026-01-05.
- [19] NOAA National Severe Storms Laboratory (NSSL). 2026. MRMS Operational Product Tables. <https://www.nssl.noaa.gov/projects/mrms/operational/tables.php>. Accessed 2026-01-05.
- [20] Open Climate Fix and Jacob Prince-Bieker. 2026. *Open Climate Fix Implementation of Deep Generative Model of Radar precipitation forecasting model*. https://github.com/openclimatefix/skillful_nowcasting
- [21] Rachel Prudden, Samantha Adams, Dmitry Kangin, Niall Robinson, Suman Ravuri, Shakir Mohamed, and Alberto Arribas. 2020. A review of radar-based nowcasting of precipitation and applicable machine learning techniques. *arXiv preprint arXiv:2005.04988* (2020).
- [22] Seppo Pulkkinen, Daniele Nerini, Andrés A. Pérez Hortal, Carlos Velasco-Forero, Alan Seed, Urs Germann, and Loris Foresti. 2019. Pysteps: An open-source Python library for probabilistic precipitation nowcasting (v1.0). *Geoscientific Model Development* 12, 10 (2019), 4185–4219. doi:10.5194/gmd-12-4185-2019
- [23] Suman Ravuri et al. 2021. Skillful precipitation nowcasting using deep generative models of radar. *Nature* 597 (2021), 672–677. doi:10.1038/s41586-021-03854-z
- [24] Registry of Open Data on AWS. 2026. NOAA Multi-Radar Multi-Sensor (MRMS) System. <https://registry.opendata.aws/noaa-mrms-pds/>. Accessed 2026-01-05.
- [25] Javier Sainz-Pardo Díaz, Miguel Castrillo, Jan Bartok, et al. 2024. Personalized federated learning for improving radar-based precipitation nowcasting on heterogeneous areas. *Earth Science Informatics* 17 (2024), 5561–5584. doi:10.1007/s12145-024-01438-9
- [26] Alan W. Seed, Clive E. Pierce, and Katie Norman. 2013. Formulation and evaluation of a scale decomposition-based stochastic precipitation nowcast scheme. *Water Resources Research* 49, 10 (2013), 6624–6641. doi:10.1002/wrcr.20536
- [27] Xingjian Shi, Zhourong Chen, Hao Wang, Dit-Yan Yeung, Wai-Kin Wong, and Wang-Chun Woo. 2015. Convolutional LSTM Network: A Machine Learning Approach for Precipitation Nowcasting. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [28] Xingjian Shi, Zhihan Gao, Leonard Lausen, Hao Wang, Dit-Yan Yeung, Wai-Kin Wong, and Wang-Chun Woo. 2017. Deep Learning for Precipitation Nowcasting: A Benchmark and a New Model. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [29] H. Soh, L. Goodrich, B. Brown, R. Bullock, J. Halley Gotway, K. Newman, J. Opatz, T. Jensen, and J. Prestopnik. 2026. The MET Version 12.2.1 User’s Guide. Developmental Testbed Center. Available at <https://github.com/dtcenter/MET/releases>.
- [30] University of Oklahoma Integrated Robust Assured Data Services. 2019. IRADS Frequently Asked Questions. <https://www.ou.edu/irads/faq.html>. Updated: May 7, 2019; Accessed: Apr. 28, 2026.
- [31] Zhe Xu, Aleksei Rutkovskii, Brenda J. Philips, and Michael H. Zink. 2025. Using Generative AI Models for High Resolution Precipitation Forecasts: A Case Study using X-band Radar Data. In *105th Annual AMS Meeting 2025 (American Meteorological Society Meeting Abstracts, Vol. 105)*. Article 457171, 457171 pages.
- [32] Jian Zhang, Kenneth Howard, Carrie Langston, et al. 2016. MRMS Quantitative Precipitation Estimation: Initial Operating Capabilities. *Bulletin of the American Meteorological Society* 97, 4 (2016), 621–637. doi:10.1175/BAMS-D-14-00174.1
- [33] Tongtian Zhu, Fengxiang He, Kaixuan Chen, Mingli Song, and Dacheng Tao. 2023. Decentralized SGD and Average-direction SAM are Asymptotically Equivalent. In *Proceedings of the 40th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 202)*, Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (Eds.). PMLR, 43005–43036. <https://proceedings.mlr.press/v202/zhu23e.html>